

Homework (Habanero)

- Q1.** A patient's medication dosage is 25. If the dosage is increased by 15%, what is the new dosage?
(A) 25
(B) 28.75
(C) 30
(D) 32.5
(E) 35
- Q2.** The population of a certain cell culture is 500 cells. If the population decreases by 8% in one hour, how many cells are left?
(A) 450 cells
(B) 460 cells
(C) 465 cells
(D) 475 cells
(E) 480 cells
- Q3.** A medical test has a 92% success rate. If 1200 tests are conducted, how many are expected to be successful?
(A) 1000
(B) 1056
(C) 1104
(D) 1105
(E) 1200
- Q4.** The price of a medical device is \$850. If a 12% discount is applied, what is the new price?
(A) \$750
(B) \$748
(C) \$750.80
(D) \$747.00
(E) \$750.50
- Q5.** A certain medicine is effective in 85% of cases. If 200 patients are treated, how many are expected to respond to the treatment?
(A) 150
(B) 160
(C) 165
(D) 170
(E) 180
- Q6.** The concentration of a solution is 20%. If 50 of the solution contains 10 of the solute, what is the total volume of the solution?
(A) 50
(A) 100
(A) 200
(A) 250
(A) 500
- Q7.** A doctor's office has 2000 patients. If 18% of them are diabetic, how many diabetic patients does the office have?
(A) 350
(B) 360
(C) 365
(D) 370
(E) 380
- Q8.** A patient is taking 5 of a certain medication daily. If the dosage is increased by 20% every week for two weeks, what is the new dosage?
(A) 6
(B) 6.5
(C) 7
(D) 7.2
(E) 8.2

Open Ended Questions (FRQ)

- Q9.** A certain medication has a 95% success rate in treating a particular disease. If 500 patients are treated with this medication and the treatment costs \$1200 per patient, what is the total cost of treatment for all patients and how many patients are expected to be cured?
- Q10.** A hospital has 1500 staff members, including doctors, nurses, and administrative staff. If 22% are doctors and 18% are nurses, what percentage of the staff are administrative staff? Show your calculations and explain your reasoning.

Chef's Tips

- Read each question carefully before selecting an answer.
- Use a calculator to check your calculations.
- Check your units and ensure that they are consistent.